Project Report Format

1. PROJECT TITLE

Quotes Recommendation Chatbot Using NLP.

2. PROJECT DESCRIPTION

 System Overview:
A chatbot that recommends motivational, inspirational, and topic-based quotes using Natural Language Processing (NLP) and intent classification.

- **Problem Domain:**
  Users struggle to find quotes that match their mood or intent quickly. Existing platforms lack personalization and emotional relevance.
- **Core Technology Used:**
  RASA NLU, Python, Flask, HTML/CSS, JavaScript.
- **Key Capabilities:**
  - Mood-based quote recommendation
  - Intent classification
  - Interactive web UI
  - Custom actions for quote categories
- **Target Users:**
  Students, professionals, and social media users seeking emotional support or inspiration.
- **Real-world Relevance:**
  Helps users feel motivated and positive during stressful moments through personalized content.

3. APPLICATION SCENARIOS / USE CASES

 End-user usage: Chatbot for daily motivation.

- **Educational usage:** Students use it for emotional upliftment.
- **Social media usage:** Users share quotes directly from the bot.
- **Enterprise usage:** Can be integrated into wellness platforms.

4. TECHNICAL ARCHITECTURE OVERVIEW

 Frontend Layer: Flask-based UI with avatars and suggestion buttons.

- **Backend Layer:** RASA server with custom actions.
- **Database Layer:** Static quote datasets (JSON/YAML).
- **External APIs:** None used currently.
- **Deployment Environment:** Localhost; future deployment on Heroku/AWS.

Explain:

- Frontend layer
- Backend layer
- Database layer (if applicable)
- External APIs / Services
- Deployment environment

5. PREREQUISITES

5.1 Software Requirements

- **Python 3.8+**

- **RASA**
- **Flask**
- **VS Code**

5.2 Libraries / Frameworks / Dependencies

- **rasa**
- **flask**
- **pytest**
- **numpy**
- **scikit-learn**

5.3 Hardware Requirements

- **Standard PC/laptop**
- **No external sensors required**

6. PRIOR KNOWLEDGE REQUIRED

- Python programming
- NLP basics
- RASA framework
- Flask web development
- Git/GitHub usage

7. PROJECT OBJECTIVES

- **Technical:** Build a chatbot that understands user mood and intent.
  - **Performance:** Fast response time, accurate quote matching.
  - **Deployment:** Host on web platform for mentor review.
  - **Learning Outcomes:** NLP implementation, chatbot design, UI integration.

8. SYSTEM WORKFLOW

- User interacts via web UI
  1. Input is sent to RASA server
  2. Intent is classified
  3. Custom action fetches relevant quote
  4. Quote is returned to UI
  5. User receives response

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

- **Problem Definition:** Lack of personalized quote recommendation tools.
  - **Functional Requirements:** Intent detection, quote retrieval, UI interaction.
  - **Non-Functional Requirements:** Fast response, scalable design.
  - **System Design Decisions:** RASA for NLP, Flask for UI.
  - **Technology Stack Justification:** RASA is modular and open-source; Flask is lightweight.

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

- **Development Setup:** Python virtual environment
  - **Dependency Installation:** pip install -r requirements.txt
  - **Project Structure:**

```
project_root/
├── actions/
├── data/
├── models/
├── tests/
├── web/
├── domain.yml
├── config.yml
└── requirements.tx
```

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

11.1 Feature 1 – Intent Classification

- Uses RASA NLU to detect user mood (e.g., motivation, love, success).

11.2 Feature 2 – Quote Retrieval

- Custom actions fetch quotes from predefined datasets.

11.3 Feature 3 – Web UI

- Flask-based interface with avatars, typing spinner, and suggestion buttons.

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

- **Integrated RASA with Flask**
- **Optimized quote matching logic**
- **Added error handling for unknown intents**

13. MILESTONE 5: TESTING & VALIDATION

Testing and validation were critical to ensure the chatbot functions reliably and delivers accurate, mood-based quote recommendations. The process involved multiple layers of testing:

- **Test Case Design:**
  Specific test cases were created to evaluate the chatbot's ability to detect user intent and respond with appropriate quotes. For example, when a user inputs "I feel sad," the bot should return a motivational quote. Similarly, inputs like "Tell me something funny" should trigger humorous responses.
- **Unit Testing:**
  Individual components such as the feedback module and custom actions were tested using pytest. This ensured that each function performed as expected in isolation.
- **Integration Testing:**
  The interaction between the RASA server and the Flask-based web interface was tested to confirm smooth data flow and response delivery. This included verifying that user inputs were correctly routed to the NLP engine and that the resulting quotes were displayed properly in the UI.
- **User Acceptance Testing (UAT):**
  The chatbot was tested by a small group of users who provided feedback on usability, relevance of quotes, and emotional impact. Their responses confirmed that the bot was intuitive and effective in delivering personalized content.
- **Performance Metrics:**
  The system was evaluated for speed and accuracy.
- **Response Time:** Less than 1 second per query.
- **Intent Classification Accuracy:** Approximately 92%.
- **Precision and Recall:** High consistency in matching user mood to quote category.
- **F1 Score:** Balanced performance across all quote types.

This milestone confirmed that the chatbot meets its functional and emotional goals, providing users with timely, relevant, and uplifting quotes through a seamless conversational interface.

14. DEPLOYMENT

- **Architecture: Localhost with RASA and Flask**
  - **Hosting Platform: Planned for Heroku**
  - **Deployment Steps:**
  - **Run RASA server**
  - **Run Flask app**
  - **Access via localhost:8080**

15. PROJECT STRUCTURE

Present clean folder structure:

```
project_root/
├── actions/
├── data/
├── models/
├── tests/
├── web/
├── domain.yml
├── config.yml
└── requirements.txt
```

16. RESULTS

- **Functional chatbot with mood-based quote delivery**
  - **Screenshots available in README.md**
  - **Performance benchmarks met**

17. ADVANTAGES & LIMITATIONS

- **Advantages: Personalized, fast, easy to use**
  - **Limitations: Limited quote categories, no live deployment yet**

18. FUTURE ENHANCEMENTS

- **Add more quote categories**
  - **Deploy on cloud**
  - **Add voice input**
  - **Mobile app integration**

19. CONCLUSION

- **Successfully built a personalized quote chatbot**
  - **Learned NLP, RASA, and web integration**
  - **Ready for mentor review and certification**

20. APPENDIX

- **Source Code: [GitHub Repo]**
- **Configuration Files: domain.yml, config.yml**
- **Dataset: Included in data/ folder**
- **Live Demo: Coming soon**

Would you like me to help you convert this into a professional PDF or DOCX format for submission? I can also add diagrams, screenshots, and polish the layout.